

E_8 Cartan matrix (Gram)

2	-1	0	0	0	0	0	0
-1	2	-1	0	0	0	0	0
0	-1	2	-1	0	0	0	0
0	0	-1	2	-1	0	0	0
0	0	0	-1	2	-1	0	-1
0	0	0	0	-1	2	-1	0
0	0	0	0	0	-1	2	0
0	0	0	0	-1	0	0	2

its inverse (integer!)

2	3	4	5	6	4	2	3
3	6	8	10	12	8	4	6
4	8	12	15	18	12	6	9
5	10	15	20	24	16	8	12
6	12	18	24	30	20	10	15
4	8	12	16	20	14	7	10
2	4	6	8	10	7	4	5
3	6	9	12	15	10	5	8

$$H^2(K3, \mathbb{Z}) \cong 3U \oplus 2E_8(-1), \text{ signature } (3, 19)$$

